How BibTeX entry variants for preprints render under REVTeX

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This document compares how different ways of writing the *same* references in a `.bib` file render in the bibliography of a typical REVTeX document (`revtex4-2` with the default `apsrev4-2` bibliography style, `hyperref` loaded). The references are an arXiv *preprint-only* paper (Wilhelm et al., arXiv:2003.10132), a HAL preprint-only paper (Turinici, HAL:hal-00640217), and a *published* paper for which a preprint exists (Koch et al., EPJ Quantum Technol. 9, 19 (2022); arXiv:2205.12110). The variants are the forms that `bibdeskparser` stores and exports, arXiv's own BibTeX export, and two instructive anti-patterns. Each variant shows its `.bib` source, followed by the citation it produces; compare the numbered entries in the bibliography at the end. This document is part of the `bibdeskparser` documentation; see the "Preprints" page there for the discussion.

I. PREPRINT-ONLY ENTRY (ARXIV)

A. P1: The stored form (`@unpublished` with pseudo-journal, eprint fields, DOI, and status note) [1]

How `bibdeskparser` (and BibDesk) store a preprint-only entry, and what any full export writes. This is the form to copy verbatim into an external `.bib` file: BibTeX ignores the `journal` field of an `@unpublished` entry, so the pseudo-journal (which serves BibDesk's table view and `bibdeskparser`'s own rendering) is invisible here. The `note` records the publication status ("preprint only", "submitted to Phys. Rev. A", "lecture notes"); `bibdeskparser` never fills it in automatically – a missing note is the signal to do so by hand.

```
@unpublished{WilhelmStored,
    Author = {Wilhelm, Frank K. and ...},
    Title = {An introduction into optimal control for quantum technologies},
    Journal = {arXiv:2003.10132},
    Eprint = {2003.10132},
    Archiveprefix = {arXiv},
    Primaryclass = {quant-ph},
    Doi = {10.48550/arxiv.2003.10132},
    Note = {lecture notes},
    Year = {2020},
}
```

B. P2: The minimal `--preprint unpublished` export (the default) [2]

What `bibdeskparser export --minimal` produces: as P1, without the pseudo-journal. For an entry without a stored note, the required `note` is synthesized as "preprint" (in minimal exports only).

```
@unpublished{WilhelmUnpublished,
    ... same fields as P1, without Journal ...
}
```

C. P3: The minimal `--preprint misc` export [3]

The same structured fields as a `@misc` entry (`@misc` has no required fields, so a note is kept but never synthesized).

```
@misc{WilhelmMisc,
    ... same fields as P2, as @misc ...
}
```

D. P4: The minimal `--preprint article` export [4]

The pseudo-journal in the journal position, hyperlinked via the DOI-resolver address in `url`; no eprint/doi fields. This is the form for classic styles (`plain`, `unsrt`, `IEEEtran`, ...), which would silently drop an `eprint`.

```
@article{WilhelmArticle,
    Author = {Wilhelm, Frank K. and ...},
    Title = {An introduction into optimal control for quantum technologies},
    Journal = {arXiv:2003.10132},
    Url = {https://doi.org/10.48550/arxiv.2003.10132},
    Note = {lecture notes},
    Year = {2020},
}
```

E. P5: arXiv's own BibTeX export (`@misc`) [5]

Verbatim what the "Export BibTeX citation" button on the arXiv abstract page produces (`bibdeskparser import` accepts this form and normalizes it to P1, deriving the DOI; only the `note` is left to fill in by hand).

```
@misc{WilhelmArxivExport,
      title={An introduction into optimal control for quantum technologies},
      author={Frank K. Wilhelm and Susanna Kirchhoff and Shai Machnes
              and Nicolas Wittler and Dominique Sugny},
      year={2020},
      eprint={2003.10132},
      archivePrefix={arXiv},
      primaryClass={quant-ph},
      url={https://arxiv.org/abs/2003.10132},
}
```

F. P6: Anti-pattern – the legacy `@article` storage form, copied verbatim [6]

The same fields as P1, but on an `@article` (the recognized legacy storage form). Copied *verbatim* into a REVTeX document instead of going through `export`, the journal field is no longer ignored, and the identifier renders three times.

```
@article{WilhelmLegacyArticle,
    ... same fields as P1, as @article, without Note ...
}
```

G. P7: Anti-pattern – `@article` with pseudo-journal and a `doi` field [7]

Why the `--preprint article` export (P4) writes the DOI as a `url` instead of a `doi` field: for an `@article` without `volume`/`pages`, `apsrev4-x` prints the DOI as visible text instead of using it as a link target.

```
@article{WilhelmArticleDoi,
    Author = {Wilhelm, Frank K. and ...},
    Title = {An introduction into optimal control for quantum technologies},
    Journal = {arXiv:2003.10132},
    Doi = {10.48550/arxiv.2003.10132},
    Year = {2020},
}
```

II. PREPRINT-ONLY ENTRY (HAL) AND THE `ARCHIVE` FIELD

REVTeX's eprint machinery builds the link text from `archivePrefix` and `eprint`, but its link *target* as `<archive>/<eprint>`, where the `archive` field defaults to arXiv's `https://arxiv.org/abs`. For a non-arXiv preprint, the default is wrong;

bibdeskparser exports therefore emit the archive field with the correct link base (derived from the archive's URL template in the preprint_archives configuration).

A. H1: The minimal --preprint unpublished export (the default) [8]

```
@unpublished{TuriniciUnpublished,
    Author = {Turinici, Gabriel},
    Title = {Quantum control},
    Eprint = {hal-00640217},
    Archiveprefix = {HAL},
    Archive = {https://hal.science},
    Url = {https://hal.science/hal-00640217},
    Note = {published in the Encyclopedia of Applied and
            Computational Mathematics, Springer, 2015,
            \url{https://doi.org/10.1007/978-3-540-70529-1_269}},
    Year = {2012},
}
```

B. H2: The minimal --preprint misc export [9]

```
@misc{TuriniciMisc,
    ... same fields as H1, as @misc ...
}
```

C. H3: The minimal --preprint article export [10]

```
@article{TuriniciArticle,
    Author = {Turinici, Gabriel},
    Title = {Quantum control},
    Journal = {HAL:hal-00640217},
    Url = {https://hal.science/hal-00640217},
    Note = {published in the Encyclopedia of Applied and
            Computational Mathematics, Springer, 2015,
            \url{https://doi.org/10.1007/978-3-540-70529-1_269}},
    Year = {2012},
}
```

D. H4: Anti-pattern – as H1, but without the archive field [11]

What every eprint-carrying form renders like for a non-arXiv archive *without* the archive field: the link points at arxiv.org/abs/hal-00640217.

```
@unpublished{TuriniciNoArchive,
    ... same fields as H1, without Archive ...
}
```

III. PUBLISHED PAPER WITH A PREPRINT

A. J1: Journal reference with the structured eprint fields [12]

How bibdeskparser stores a published paper with a preprint, and what both full and minimal exports write.

```
@article{KochFull,
    Author = {Koch, Christiane P. and ...},
    Title = {Quantum optimal control in quantum technologies. ...},
    Journal = epjqt,
```

```
    Eprint = {2205.12110},
    Archiveprefix = {arXiv},
    Doi = {10.1140/epjqt/s40507-022-00138-x},
    Pages = {19},
    Volume = {9},
    Year = {2022},
}
```

B. J2: The same entry without the eprint fields [13]

What the bibliography loses by dropping `eprint`/`archiveprefix`: the free-copy link for readers behind a paywall.

```
@article{KochNoEprint,
    ... as J1, without Eprint/Archiveprefix ...
}
```

C. J3: A published paper with a HAL preprint, as exported [14]

Full and minimal exports of published entries also append the `archive` link base when the `eprint` names a non-arXiv archive.

```
@article{SauvagePRXQ,
    Author = {Sauvage, Fr{\'e}d{\'e}ric and Mintert, Florian},
    Title = {Optimal Quantum Control with Poor Statistics},
    Journal = {PRX Quantum},
    Year = {2020},
    Doi = {10.1103/prxquantum.1.020322},
    Pages = {020322},
    Volume = {1},
    Number = {2},
    Eprint = {hal-03612955},
    Archiveprefix = {HAL},
    Archive = {https://hal.science},
}
```

IV. OBSERVATIONS

In the table below, $\langle\ldots\rangle$ marks the extent of a hyperlink in the rendered entry (follow the links in the bibliography itself to verify), and "id" abbreviates the identifier of the respective reference. Authors and title precede each rendering.

P1 [1]	$\langle$arXiv:id [quant-ph]$\rangle_{\text{arXiv}}$ (2020), lecture notes — identifier in the venue position, category tag, status
P2 [2]	identical to P1 (the pseudo-journal is ignored)
P3 [3]	$\langle$title$\rangle_{\text{DOI}}$ (2020), lecture notes, $\langle$arXiv:id [quant-ph]$\rangle_{\text{arXiv}}$ — DOI-linked title; eprint trails after the year
P4 [4]	$\langle$arXiv:id (2020)$\rangle_{\text{DOI}}$, lecture notes — venue-position identifier under *every* style
P5 [5]	$\langle$title$\rangle_{\text{arXiv}}$ (2020), $\langle$arXiv:id [quant-ph]$\rangle_{\text{arXiv}}$ — like P3, but no DOI and no status
P6 [6]	arXiv:id $\langle$10.48550/arxiv.id$\rangle_{\text{DOI}}$ (2020), $\langle$arXiv:id [quant-ph]$\rangle_{\text{arXiv}}$ — identifier shown *three* times, raw DOI visible
P7 [7]	arXiv:id $\langle$10.48550/arxiv.id$\rangle_{\text{DOI}}$ (2020) — raw DOI visible next to the pseudo-journal
H1 [8]	$\langle$HAL:hal-00640217$\rangle_{\text{HAL}}$ (2012), $\langle$Note$\rangle_{\text{DOI}}$ — eprint link correct, thanks to archive
H2 [9]	$\langle$title$\rangle_{\text{HAL}}$ (2012), $\langle$Note$\rangle_{\text{DOI}}$, $\langle$HAL:hal-00640217$\rangle_{\text{HAL}}$ — all links correct
H3 [10]	$\langle$HAL:hal-00640217 (2012)$\rangle_{\text{HAL}}$, $\langle$Note$\rangle_{\text{DOI}}$ — both links correct
H4 [11]	$\langle$HAL:hal-00640217$\rangle_{\text{broken}}$ (2012), $\langle$Note$\rangle_{\text{DOI}}$ — without archive, the eprint link points at arxiv.org
J1 [12]	$\langle$EPJ Quantum Technol. **9**, 19 (2022)$\rangle_{\text{DOI}}$, $\langle$arXiv:id$\rangle_{\text{arXiv}}$ — journal reference *and* free-copy link
J2 [13]	as J1, but *without* the arXiv link
J3 [14]	$\langle$PRX Quantum **1**, 020322 (2020)$\rangle_{\text{DOI}}$, $\langle$HAL:hal-03612955$\rangle_{\text{HAL}}$ — the free-copy link is correct, thanks to archive

Key observations, specific to apsrev4-2 (2019-01-14) with hyperref:

1. The stored @unpublished form (P1) can be copied verbatim into a document: it renders exactly like the minimal export (P2), because BibTeX ignores the journal field of an @unpublished entry. The eprint renders hyperlinked in the venue position, with the category tag and the status note – the best REVTeX rendering of a preprint. The legacy @article storage form (P6) can *not* be copied verbatim; bibdeskparser export normalizes either stored form.

2. note is a *required* field of @unpublished: BibTeX warns "empty note" for every entry without one. Recording the publication status there ("preprint only", "submitted to Phys. Rev. A", "lecture notes") is therefore doubly useful; minimal exports synthesize "preprint" when an entry stores none.

3. The three export forms are complementary. unpublished (P1/P2, H1) and misc (P3, H2) rely on the style rendering the eprint field — which REVTeX, elsarticle, and biblatex do, but plain, unsrt, abbrv, and IEEEtran do *not* (under those, only the note text survives, and the identifier is lost). article (P4, H3) renders the identifier in the journal position under *every* style, at the cost of the native eprint formatting (no category tag).

4. For an @article without volume/pages, the doi field is printed as *visible text* (P6, P7) rather than becoming the link target of the journal reference, as it does when volume/pages are present (J1, J3). The article export form therefore writes the DOI as a url, which wraps the journal name and year in the hyperlink (P4, H3).

5. The link target of REVTeX's rendered eprint is <archive>/<eprint>, with the archive field defaulting to https://arxiv.org/abs. For a non-arXiv archive that default is a broken link (H4); emitting the correct base in archive fixes the eprint link everywhere it appears (H1, H2, J3). The url field cannot compensate on @unpublished, where it is ignored.

6. Keeping the structured eprint fields on a *published* entry costs nothing and adds the hyperlinked identifier after the journal reference (J1/J3 vs. J2) — the link that stays accessible to readers without a journal subscription. bibdeskparser's minimal export therefore includes them (and every export adds the archive base where needed, J3).

[1] F. K. Wilhelm, S. Kirchhoff, S. Machnes, N. Wittler, and D. Sugny, An introduction into optimal control for quantum technologies, arXiv:2003.10132 [quant-ph] (2020), lecture notes.

[2] F. K. Wilhelm, S. Kirchhoff, S. Machnes, N. Wittler, and D. Sugny, An introduction into optimal control for quantum technologies, arXiv:2003.10132 [quant-ph] (2020), lecture notes.

[3] F. K. Wilhelm, S. Kirchhoff, S. Machnes, N. Wittler, and D. Sugny, An introduction into optimal control for quantum technologies (2020), lecture notes, arXiv:2003.10132 [quant-ph].

[4] F. K. Wilhelm, S. Kirchhoff, S. Machnes, N. Wittler, and D. Sugny, An introduction into optimal control for quantum technologies, arXiv:2003.10132 (2020), lecture notes.

[5] F. K. Wilhelm, S. Kirchhoff, S. Machnes, N. Wittler, and D. Sugny, An introduction into optimal control for quantum technologies (2020), arXiv:2003.10132 [quant-ph].

[6] F. K. Wilhelm, S. Kirchhoff, S. Machnes, N. Wittler, and D. Sugny, An introduction into optimal control for quantum technologies, arXiv:2003.10132 10.48550/arxiv.2003.10132 (2020), arXiv:2003.10132 [quant-ph].

[7] F. K. Wilhelm, S. Kirchhoff, S. Machnes, N. Wittler, and D. Sugny, An introduction into optimal control for quantum technologies, arXiv:2003.10132 10.48550/arxiv.2003.10132 (2020).

[8] G. Turinici, Quantum control, HAL:hal-00640217 (2012), published in the Encyclopedia of Applied and Computational Mathematics, Springer, 2015, `https://doi.org/10.1007/978-3-540-70529-1_269`.

[9] G. Turinici, Quantum control (2012), published in the Encyclopedia of Applied and Computational Mathematics, Springer, 2015, `https://doi.org/10.1007/978-3-540-70529-1_269`, HAL:hal-00640217.

[10] G. Turinici, Quantum control, HAL:hal-00640217 (2012), published in the Encyclopedia of Applied and Computational Mathematics, Springer, 2015, `https://doi.org/10.1007/978-3-540-70529-1_269`.

[11] G. Turinici, Quantum control, HAL:hal-00640217 (2012), published in the Encyclopedia of Applied and Computational Mathematics, Springer, 2015, `https://doi.org/10.1007/978-3-540-70529-1_269`.

[12] C. P. Koch, U. Boscain, T. Calarco, G. Dirr, S. Filipp, S. J. Glaser, R. Kosloff, S. Montangero, T. Schulte-Herbrüggen, D. Sugny, and F. K. Wilhelm, Quantum optimal control in quantum technologies. strategic report on current status, visions and goals for research in Europe, EPJ Quantum Technol. **9**, 19 (2022), arXiv:2205.12110.

[13] C. P. Koch, U. Boscain, T. Calarco, G. Dirr, S. Filipp, S. J. Glaser, R. Kosloff, S. Montangero, T. Schulte-Herbrüggen, D. Sugny, and F. K. Wilhelm, Quantum optimal control in quantum technologies. strategic report on current status, visions and goals for research in Europe, EPJ Quantum Technol. **9**, 19 (2022).

[14] F. Sauvage and F. Mintert, Optimal quantum control with poor statistics, PRX Quantum **1**, 020322 (2020), HAL:hal-03612955.